Nima Daneshvarnejad, Ph.D.

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Education

2022 – 2026 **University of Southern California, Los Angeles, CA**

8/2026 Ph.D. in Energy and Petroleum Engineering
Thesis: *IoT-Enabled Static Flux Chamber Monitoring System for Methane Detection from Plugged and Unplugged Wells*
Advisor: Iraj Ershaghi, Donald L. Paul
Funding: USC Viterbi-CVX UPP Energy Resources Doctoral Fellowship, USC Graduate School Fellowship

5/2022 M.S. in Petroleum Engineering
Funding: USC Viterbi-CVX UPP Energy Resources M.S. Scholarship

2013 – 2017 **Sharif University of Technology, Tehran, Iran**

5/2017 B.S. in Petroleum Engineering

Publications

1. Daneshvarnejad, N., Gandhi, Y.P., Cho, Y., Kalia, R., Farhadi, S., Paul, D., and Ershaghi, I. 2026. "Challenges in Designing a New Smart Static Methane Detection and Monitoring Chamber: Application to Plugged and Unplugged Abandoned Wells." *Environments* 13, no. 7: 411. https://doi.org/10.3390/environments13070411
2. Daneshvarnejad, N., Gandhi, Y.P., Kalia, R., Cho, Y., Paul, D., and Ershaghi, I. "AI-Driven Computational Fluid Dynamic Simulations and Experiments to Predict Methane Emission: Applications to Idle and Abandoned Wells." Paper presented at the SPE Western Regional Meeting, Garden Grove, California, USA, April 2025. doi: https://doi.org/10.2118/224153-MS
3. Daneshvarnejad, N., Khodaparast, P., Cho, Y., Paul, D., Hauser, M., Farhadi, S., and Ershaghi, I. "Monitoring Low and Intermittent Methane Emission from Orphaned and Idle Wells." Paper presented at the SPE Western Regional Meeting, Palo Alto, California, USA, April 2024. doi: https://doi.org/10.2118/218839-MS
4. Rehak, M., Warburton, D., Daneshvarnejad, N., and Ershaghi, I. "The Integrity of Idle and Abandoned Wells in California." Paper presented at the SPE Western Regional Meeting, Palo Alto, California, USA, April 2024. doi: https://doi.org/10.2118/218887-MS
5. Daneshvarnejad, N., Khodaparast, P., Cho, Y., Paul, D., Farhadi, S., and Ershaghi, I. "A Method for Detection of Methane Leaks from Idle and Orphaned Wells Using High-Precision Sensor and a Ventilation Canopy." Paper presented at the SPE Canadian Energy Technology Conference and Exhibition, Calgary, Alberta, Canada, March 2024. doi: https://doi.org/10.2118/218022-MS

Honors and Awards

2026	Second Place in the Ph.D. Division for the SPE International Student Paper Contest
2026	Doctoral Award of Excellence (Association of Professors and Scholars of Iranian Heritage)
2026	Dr. Ali Daneshy Award (USC-ECET National Academy of Engineering Annual Summit)
2026	Best Doctoral Student Paper Award (Society of Petroleum Engineers, Western Regional)
2025	USC Viterbi School of Engineering “Best Research Assistant (RA) Student” Award
2023-2026	Graduate Student Award of Excellence (Society of Petroleum Engineers, LA Basin)
2023	Chevron Energy Graduate Student Fellowship (Ph.D.)
2022	USC Viterbi School of Engineering Student Fellowship
2021	Chevron Energy Graduate Student Fellowship (M.S.)

Research Mentoring

08/2024 – 05/2026	**M. Nasseri, Ph.D. Student, Petroleum Engineering** University of Southern California (Ershaghi Lab)
03/2025 – 12/2025	**B. Khan, M.S. Student, Petroleum Engineering** University of Southern California (Ershaghi Lab)
01/2024 – 12/2024	**D. Gayen, M.S. Student, Chemical Engineering** University of Southern California (Ershaghi Lab)
08/2022 – 08/2024	**Y. P. Gandhi, M.S. Student, Computer Science** University of Southern California (Ershaghi Lab)
01/2023 – 10/2023	**J. Dowlani, M.S. Student, Petroleum Engineering** University of Southern California (Ershaghi Lab)
05/2022 – 12/2022	**M. Rehak, B.S. Student, Chemical Engineering** Graduate Research Intern, University of Southern California (USC CiSoft Lab)

Teaching Experience

Fall 2022	**Teaching Assistant, University of Southern California** PTE 461 – Formation Evaluation (Graduate Level)
Spring 2022	**Teaching Assistant, University of Southern California** PTE 555 – Well Completion, Stimulation, and Damage Control (Graduate Level)
Summer 2014-2016	**Teaching Assistant, Sharif University of Technology** Oil and Gas Properties Laboratory Section (Undergraduate Level)

Professional Memberships

2018 – 2026	American Institute of Chemical Engineers (AIChE)
2013 – 2026	Society of Petroleum Engineers (SPE)
2023 – 2026	American Geophysical Union (AGU)
2023 – 2026	American Association of Petroleum Geologists (AAPG)
2024 – 2026	American Meteorological Society (AMS)

Invited Talks

2026	2nd Annual Orphan, Idle and Marginal Wells California Conference Presenter and Panelist *Using Carbon Markets to Scale Monitoring and Remediation of Legacy Wells in California*
2026	4th AAPG Orphan, Abandoned, Idle, and Marginal Wells Conference *From Leak Rate to Leak Profile: AI-Driven Monitoring and Predictive Modeling of Methane Emissions from Orphaned and Abandoned Wells*
2026	U.S. Geological Survey Geology, Energy & Minerals (GEM) Science Center *IoT-Enabled Static Flux Chamber Monitoring System for Methane Detection from Plugged and Unplugged Abandoned Wells*
2025	McGill University, Laboratory of Professor Mary Kang *From Leak Rate to Leak Profile: AI-Driven Monitoring and Predictive Modeling of Methane Emissions from Orphaned and Abandoned Wells*
2025	U.S. Department of Energy, National Energy Technology Laboratory *Methane Emission Monitoring and Mitigation*

2025 San Joaquin Well Logging Society
Well Abandonment and Misconception

2025 3rd AAPG Orphan, Abandoned, Idle, and Marginal Wells Conference
The Importance of Intermittent Well Leaks in Voluntary Carbon Credit Markets

Skills

Core Programming: Python [advanced], MATLAB [advanced], R [advanced], Java [intermediate], Julia [intermediate], High-Performance Computing [Slurm], Version Control [Git, GitHub]

Data Science / ML: Keras, TensorFlow, PyTorch, scikit-learn, Liquid-time constant networks, Fourier neural operators (FNO), Physics-Informed Neural Networks

Numerical and Computational Fluid Dynamics Modeling: COMSOL (laminar fluid flow, turbulent fluid flow, Transport of Concentrated Species), LiveLink for MATLAB

Air Quality Monitoring & GIS: AERMOD, HARP2 Risk Assessment Standalone Tool (RAST), ArcGIS, CMAQ, SPECIATE

Web Dev: HTML, Netlify

Professional Experience

2024-2026 **Industry Outreach and Air Quality Researcher at CrudeCast**
Led air-quality research with the Energy Policy Institute at the University of Chicago, Dr. Daniel Westervelt, and *Wasaru* using low-cost sensor data measuring $PM_{2.5}$, PM_{10}, and VOCs from Kinshasa for seasonal analysis and air-quality forecasting. Developed an LLM-based tool for public air-quality information in the Democratic Republic of Congo.

References

Name	Affiliation	Contact Information
Dr. Iraj Ershaghi	University of Southern California	ershaghi@usc.edu 310-991-1337
Dr. Donald Paul	University of Southern California	donaldpa@usc.edu 213-740-1076
Dr. Rajiv Kalia	University of Southern California	rkalia@usc.edu 213-821-2658
Tedrik Hayrapetian	South Coast AQMD	ted.hayrapetian@gmail.com 818-797-8681
Dr. Cyrus Ashayeri	Beyond Limits, CrudeCast	cyrus.ashayeri@gmail.com 310-570-5956